V2.0 NTA advies apartments

V2.0 NTA advies

Description

This endpoint returns the full calculation for apartments. It gives the results of the current building, as well as the results after measures applied. The input is the current building. Based on criteria and filters, the tool calculates the optimal output.

Endpoint: /v2.0

Api input json structure ->/V2.0 (NTA advies Multi)

```
{
 "ConstructionYearCategory": int,
 "Subtype": int,
 "LivingArea": float,
 "NumberOfStories": int,
 "BackFacade": int,
 "WallInsulation": int,
 "FloorInsulation": int,
 "RoofInsulation": int,
 "GlassLivingArea": int,
 "Installation": int,
 "ShowerHeatRecovery": int,
 "Cooling": int,
 "Ventilation": int,
 "ElectricCooking": int,
 "Residents": int,
 "GasConsumption": float,
 "HeatConsumption": float,
 "ElectricityConsumption": float,
 "ElectricityGeneration": float,
 "MinimumPanels": int,
 "SolarPanels": [
  {
    "PVTotalWattPeak": float,
    "PVArea": float,
    "SpecificWattpeak":float,
    "InstallationAngle": int,
    "Orientation": int,
    "FreeRoofArea": float,
  },
  {....}
 ],
 "Criterion": int,
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```
"TargetLabel": int,
"StrictTargetLabel": int,
"MaximumInvestment": float,
"NaturalGasChoice": int,
"HeatpumpAllowed": int,
"DistrictHeatingAllowed": int,
"MinimumInsulationHeatpump": int,
"LockedMeasures": [int],
"ExcludeSubMeasures": [int],
"ExcludingVat": int,
"CostIndicators": {
  "fixed_gas_price": float,
  "variable_gas_price": float,
  "energy_tax_gas": float,
  "fixed_electricity_price": float,
  "variable_electricity_price": float,
  "energy_tax_electricity": float,
  "fixed_heat_price": float,
  "energy_tax_heat": float,
  "variable_heat_price": float,
  "tax_credit": float,
  "feed_in_compensation": float,
  "feed_in_cost": float,
  "fixed_solar_panel_costs": float,
  "netting_percentage": float,
  "gas_yearly_price_increase": float,
  "electricity_yearly_price_increase": float,
  "external_heating_yearly_price_increase": float,
  "term": int,
  "discount_rate": float,
  "tax_rate": float,
  "custom_measure_costs": {
        “code”: str,
        “value”: float
  }
},
"ResultsPerMeasure": int,
"FloorMeasures": {
  "FloorType": int,
  "CrawlSpace": bool,
  "Material": int
},
"WallMeasures": {
  "Biobased": bool,
  "HasCavity": bool,
  "Material": int
},
"RoofMeasures": {
```

```
    "Biobased": bool,
    "Side": int,
    "ReplaceRoof": bool,
    "Material": int
  },

}
```

Database

System environment variable with full connection string
`DATABASE_URL="postgresql+psycopg2://username:password@host/dbname"`

Parameter explanation

When a parameter is not provided, the default value will be used

Fixed building parameters

Name	Description	Values	Default	Notes
ConstructionYearCategory	The period in which the house was constructed.	1 = t/m 1924 2 = from 1925 t/m 1945 3 = from 1946 t/m 1964 4 = from 1965 t/m 1974 5 = from 1975 t/m 1991 6 = from 1992 t/m 2005 7 = from 2006 t/m 2014 8 = from 2015	-	Mandatory (CYC)
SubType	The type of apartment	1 = corner ground floor 2 = in between ground floor 3 = corner mid floor 4 = in between mid floor 5 = corner top floor 6 = in between top floor 7 = entire top floor	-	Mandatory
NumberOfStories	The number of stories of the unit	1 = 1 story/building layers 2 = 2 stories/building layers	-	Mandatory
BackFacade	If the apartment has as back facade (achtergevel)	0 = no back facade 1 = back facade	-	Mandatory
LivingArea	The total living area, (gebruiksoppervlak)	Float, > 0.0 in m^2	-	Mandatory, soft limit on min and max

Current envelope and installation parameters

Name	Description	Values	Default	Notes
WallInsulation	The insulation level of the	1 = Geen	1 if CYC <=4,	

	walls.	2 = Matig 3 = Goed 4 = Zeer Goed	2 if CYC <=5, 3 if CYC <=6, 4 if CYC > 6	
FloorInsulation	The insulation level of the floor.	1 = Geen 2 = Matig 3 = Goed 4 = Zeer Goed	1 if CYC <=4, 2 if CYC <=5, 3 if CYC <=6, 4 if CYC > 6	
RoofInsulation	The insulation level of the roof.	1 = Geen 2 = Matig 3 = Goed 4 = Zeer Goed	1 if CYC <=4, 2 if CYC <=5, 3 if CYC <=6, 4 if CYC > 6	
GlassLivingArea	The type of glass in the living area.	1 = Single 2 = Double 3 = HR++ 4 = Triple	2 if CYC <=6, 3 if CYC > 6	
Installation	The type of installation used for heating and hot water	1 = Lokaal gas + geiser 2 = VR ketel + geiser 3 = VR combi ketel 4 = HR combi ketel 6 = Elektrishe lucht-water WP 7 = Stadverwarming 8 = Hybride WP 9 = Collectieve HR ketel 10 = Collectieve warmtepomp	4	
ShowerHeatRecovery	Is there a shower heat recovery system	1 = No shower heat recovery 2 = Vertical shower heat recovery system	1	
Cooling	Is there a cooling system present	1 = No cooling 2 = air conditioning unit, direct air cooling	1	
Ventilation	The type of ventilation system present	1 = Natuurlijke ventilatie 2 = Mechanische afzuiging 3 = Balansventilatie met wtw 4 = Decentrale balansventilatie	1 if CYC <=4, 2 if CYC <=7, 3 if CyC > 7	
ElectricCooking	What is used for cooking	1 = Gas stove 2 = Electric stove	2 if Installation in [6,7], Else 1	

User related parameters

Name	Description	Values	Default	Notes
Residents	Number of residents	Int > 0. Values > 6 will be set to 6	3	
GasConsumption	User specified yearly gasconsumption in m^3	Float > 0.0	calculation	
HeatConsumption	User specified yearly heat consumption in GJ.	Float > 0.0	calculation	Only with installation 7
ElectricityConsumption	The gross user specified yearly electricity consumption in KWh	Float, > 0.0	calculatoin	
ElectricityGeneration	The total yearly electricity yield of solar panels in kWh. Net usage = consumption -	Float, > 0.0	calculation	

	yield			

Solar panels

Name	Description	Values	Default	Notes
MinimumPanels	The minimum number of new panels to be placed	Int, > 0	4	
SolarPanels	A list of solar panel objects, including roof parameters	list	[]	See below

SolarPanels field

Name	Description	Values	Default	Notes
PVTotalWattPeak	The total wattpeak power of the existing system (watt piek). = PVArea * SpecificWattpeak	float, > 0	calculation	When inputting existing system: either PVTotalWattPeak or (PVArea and SpecificWattpeak) is mandatory. 0 if there is no existing system
PVArea	The total effective area of all existing solar panels combined. Standard panel size is 1.65 m^2	Float, > 0.0	1.65 m^2 per panel	When inputting existing system: either PVTotalWattPeak or (PVArea and SpecificWattpeak) is mandatory. 0 if there is no existing system
SpecificWattpeak	The wattpeak/m^2 of the panels. Standard panel size is 1.65 m^2	Float, > 0.0	250 wattpeak/m^2	When inputting existing system: either PVTotalWattPeak or (PVArea and SpecificWattpeak) is mandatory. 0 if there is no existing system
InstallationAngle	The angle of the solar panels	int> 0, <= 90	45	mandatory
Orientation	The orientation of the solar panels	0 = N 45 = NO 90 = O 135 = ZO 180 = Z 225 = ZW 270 = W 315 = NW		mandatory
FreeRoofArea	The free space on this roof surface in m^2 that can be used for additional panels.	Float > 0.0	0	

Filters

Name	Description	Values	Default	Notes
Criterion	The sorting criterion of the final result	1 = lowest investment costs 2 = lowest energy costs 3 = highest return on investment 4 = lowest beng2 (highest label)	3	

TargetLabel	The (minimum) label to be reached by the improvements	1 = A++++ 2 = A+++ 3 = A++ 4 = A+ 5 = A 6 = B 7 = C 8 = D 9 = E 10 = F 11 = G	11	
StrictTargetLabel	Are results better than target label allowed.	0 = not strict, result can be higher label 1 = strict, only results in target label	0	
MaximumInvestment	The maximum amount in euro to be invested	Float, > 0.0	999999	
NatrualGasChoice	Natural gas allowed after improvements	1 = nofilter 2 = natural gas mandatory 3 = natural gas free	1	
HeatPumpAllowed		0 = not allowed 1 = allowed	0	
DistrictHeatingAllowed		0 = not allowed 1 = allowed	0	
MinimumInsulationHeatpump	Minimum requirement on insulation before heatpump is allowed	1 = no minimum requirement 2 = envelope, including glass min 2 3 = house meets “standaard”	3	
LockedMeasures	Current parameters that are not allowed to be improved	[1] = WallInsulation [2] = FloorInsulation [3] = RoofInsulation [4] = GlassLivingArea [5] = Instalation [6] = ShowerHeatRecovery [7] = Cooling [8] = Ventilation [9] = SolarPanels [10] =ElectricCooking	[]	Add all locked measures to the list
ExcludeSubmeasures	Specific measures to be excluded from the results	[1002] = WI matig [1003] = WI goed [1004] = WI zeer goed [2002] = FI matig [2003] = FI goed [2004] = FI zeer goed [3002] = RI sloped matig [3003] = RI sloped goed [3004] = RI sloped zeer goed [4002] = Glass living dubbel [4003] = Glass living hr++ [4004] = Glass living drievoudig [5004] = HR combi [5005] = HR combi met zonneboiler [5006] = Elek warmtepomp [5008] = Hybride warmtepomp [8002] = Mechanische afzuiging [8003] = Balans wtw [8004] = Decentrale balans	[]	

Financial parameters

Name	Description	Values	Default	Notes
ExcludingVat	Prices are with or without VAT included.	0 = VAT included 1 = VAT excluded	0	Not yet fully

				implement ed
ConstIndicators	ConsIndicator object	json	{}]	See below

CostIndicators field

(standards from cbs may 2025, https://www.cbs.nl/nl-nl/cijfers/detail/85592NED

Name	**Description**	**Values**	**Default**	**Notes**
fixed_gas_price	Fixed gas cost.Transporttarief + vast leverlingstarief.	Float, > 0.0	337.89	In €/year
variable_gas_price	Variable price exc. tax. Variabel leveringstarief contractprijs	Float > 0.0	0.6595	In €/m3
energy_tax_gas	Energy tax on gas.	Float > 0.0	0.69957	In €/m3
fixed_electricity_price	Fixed electricity price. Transporttarief + vast leveringstarief	Float, > 0.0	575.76	In €/year
variable_electricity_price	Variable price exc. tax. Variabel leveringstarief contractprijs	Float, > 0.0	0.1470	In €/kWh
energy_tax_electricity	Energy tax on electricity.	Float, > 0.0	0.12286	In €/kWh
fixed_heat_price	From Vattenvall, tarifs 2025	Float, > 0.0	760.77	In €/year
energy_tax_heat	Tax on consumed external heat	Float, > 0.0	0	In €/GJ
variable_heat_price	From Vattenvall, including, rent afleverset	Float, > 0.0	43.79	In €/GJ
tax_credit	Vermindering energiebelasting	Float, > 0.0	635.19	In €/year
feed_in_compensation	Earnings per kwh fed back in the grid	Float, > 0.0	0.1331	€/kWh
feed_in_cost	Cost per kwh fed back in the grid	Float, > 0.0	0.125	€/kWh
fixed_solar_panel_cost	Yearly extra charge from electricity supplier if you have solar panels	Float, > 0.0	0.0	€/year
netting_percentage	The fraction of produced electricity that is returned to the grid (and will count towards salderingsregeling.	Float, > 0.0	0.65	
gas_yearly_price_increase	Expected percentage increase in gas price	Float, > 0.0	0.03	
electricity_yearly_price_increase	Expected percentage increase in electricity price	Float, > 0.0	0.02	
external_heating_yearly_price_increase	Expected percentage increase in external heating price	Float, > 0.0	0.03	
term	The term in years used for calculation return on investment	Int, > 0	25	year
discount_rate	Discount rate (~inflation) used in return on investment calculation	Float	0.04	
tax_rate	General VAT	float	1.21	

custom_measure_costs	{“RVO_code” value}	json	{}	To be expanded with filtereable parameters

Output parameters

Name	Description	Values	Default	Notes
ResultsPerMeasure	Whether subresults for each individual measure are shown	0 = don’t show 1 = show	0	

Measure filters

Name	Description	Values	Default	Notes
FloorMeasures	Filters applied to allowable floor measures	json		
WallMeasures	Filters applied to allowable wall measures	json		
RoofMeasures	Filters applied to allowable roof measures	json		

FloorMeasures field

Name	Description	Values	Default	Notes
FloorType	Type of floor (material)	1: wood floor 2: concrete floor	None	
CrawlSpace	Does the house have a crawlspace >isulation underside of floor possible	True: crawlspace False: no crawlspace	None	
Material	Material of the insulation	1: mineral wool 2: PIR 3: EPS 4: Resol	None	

WallMeasures field

Name	Description	Values	Default	Notes
Biobased	Use biobased materials?	True: only use biobased options False: both biobased and non-biobased	None	
HasCavity	Are cavity walls present suitable for insulation	True: cavity exists and can be filled False: no cavity walls or not suitable	None	
Material	Material of the insulation	1: mineral wool 2: PIR 3: EPS pearls 4: Wood fiber	None	

		5: PU foam		

RoofMeasures field

Name	Description	Values	Default	Notes
Biobased	Use biobased materials?	True: only use biobased options False: both biobased and non-biobased	None	
Side	Inside of outside of roof	1: only inside insulation 2: only outside insultation	None	
ReplaceRoof	Consider replacing the roof?	True: Options include ones with roof replacement False: Only options without roof replacement		
Material	Material of the insulation	1: PIR 2: Bio EPS 3: EPS 4: XPS 5: Flax 6: Wood fibre 7: Mineral wool 8: Resol	None	

Output

Json output structure

```
{
 "Warnings": [
  {
   "code": "string",
   "description": "string"
  },
  {
   "code": "01",
   "description": "string"
  }
 ],
 "Current_House": {
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  "BENG2": int,
  "Wall_Insulation": int,
  "Floor_Insulation": int,
  "Roof_Insulation": int,
  "Glass_Living_Area": int,
  "Installation": int,
  "Shower_Heat_Recovery": int,
  "Cooling": int,
  "Ventilation": int,
  “ElectricCooking”: int,
```

```
  "Solar_Panels": [
    {
      "Roof_Surface": 0,
      "Total_Watt_Peak_Capacity": int,
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      "Orientation": int
    },
    {
      "Roof_Surface": n,
      "Total_Watt_Peak_Capacity": int,
      "Inclination_Angle": int,
      "Orientation": int
    }
  ],
  "CO2_Emission": int,
  "Gross_Electricity_Use_KWh": int,
  "Solar_panel_Yield_KWh": int,
  "Gas_Use_M3": int,
  "External_Heating_Use_Gj": int,
  "Yearly_Energy_Cost": int,
  "Monthly_Energy_Cost": float
},
"Estimated_Geometric_Areas": {
  "floor": float,
  "wall": float,
  "roof":float,
  "window_living": float,
  "doors": float
},
"Resulting_House": {
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  "BENG2": int,
  "Wall_Insulation": int,
  "Floor_Insulation": int,
  "Roof_Insulation": int,
  "Glass_Living_Area": int,
  "Installation": int,
  "Shower_Heat_Recovery": int,
  "Cooling": int,
  "Ventilation": int,
  “ElectricCooking”: int,
  "Solar_Panels": [
    {
      "Roof_Surface": 0,
      "Original_Watt_Peak_Capacity": int,
     ,"Added_Watt_Peak_capacity": int,
      "Added_Panels": int,
      "Inclination_Angle": int,
```

```
        "Orientation": int
      },
      {
        "Roof_Surface": n,
        "Original_Watt_Peak_Capacity": int,
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        "Added_Panels": int,
        "Inclination_Angle": int,
        "Orientation": int
      }
    ],
    "CO2_Emission": int,
    "Gross_Electricity_Use_KWh": int,
    "Solar_panel_Yield_KWh": int,
    "Gas_Use_M3": int,
    "External_Heating_Use_Gj": int,
    "Yearly_Energy_Cost": int,
    "Monthly_Energy_Cost":int,
    "Yearly_Energy_Savings": int,
    "Montyly_Energy_Savings": int,
    "Investment_Costs": int,
    "Return_On_Investment": float,
    "CO2_Reduction": int,
    “Variable_Gas_Price”: float,
    “Variabel_Electricity_Price”: float,
    “Solarpanel_Wattpeak_per_M2”: int
  },
  "Results_per_Measure": {
       "<measure name>": {
      "Current_Value": int,
      "New_Value": int,
      “Measure_code”: [str]
      "Investment": int,
      “Electricity_saving_kwh”: int,
      “Gas_saving_m3”: int,
      "Energy_Cost_Savings": int,
      "CO2_Reduction": int,
      "Relative_BENG2_Improvement": int
      },
     {
       …
      }
   }
}
```

delta apartments

Delta

Description

With this endpoint, the user specifies both the current situation and the desired outcome.
The tool calculates the energetic performance, energy usage/costs and investment costs.

Endpoint: /delta

Api input json structure -> /delta (Multi Delta)

{
"ConstructionYearCategory": int,
"Subtype": int,
"LivingArea": float,
"NumberOfStories": int,
"BackFacade": int,
"WallInsulation": int,
"FloorInsulation": int,
"RoofInsulation": int,
"GlassLivingArea": int,
"Installation": int,
"ShowerHeatRecovery": int,
"Cooling": int,
"Ventilation": int,
"ElectricCooking": int,
"TargetWallInsulation": int,
"TargetFloorInsulation": int,
"TargetRoofInsulation": int,
"TargetGlassLivingArea": int,
"TargetInstallation": int,
"TargetShowerHeatRecovery": int,
"TargetCooling": int,
"TargetVentilation": int,
"TargetElectricCooking": int,
"Residents": int,
"SolarPanels": [
{
"PVTotalWattPeak": float,
"PVArea": float,
"SpecificWattpeak": float,
"PVTotalWattPeakTarget": float,
"PVAreaTarget": float,
"SpecificWattpeakTarget": int,
"InstallationAngle": int,

```
    "Orientation": int,
  },
  {....}
],
"ExcludingVat": int,
"CostIndicators": {
  "fixed_gas_price": float,
  "variable_gas_price":float,
  "energy_tax_gas":float,
  "fixed_electricity_price":float,
  "variable_electricity_price": float,
  "energy_tax_electricity": float,
  "fixed_heat_price": float,
  "energy_tax_heat": float,
  "variable_heat_price": float,
  "tax_credit": float,
  "feed_in_compensation": float,
  "feed_in_cost": float,
  "fixed_solar_panel_costs": float,
  "netting_percentage": float,
  "gas_yearly_price_increase": float,
  "electricity_yearly_price_increase": float,
  "external_heating_yearly_price_increase": float,
  "term": int,
  "discount_rate": float,
  "tax_rate": float,
  "custom_measure_costs": {
        “code”: str,
        “value”: float
  }
},
"ResultsPerMeasure": int,
"FloorMeasures": {
  "FloorType": int,
  "CrawlSpace": bool,
  "Material": [int]
},
"WallMeasures": {
  "Biobased": bool,
  "HasCavity": bool,
  "Material": [int]
},
"RoofMeasures": {
  "Biobased": bool,
  "Side": int,
  "ReplaceRoof": bool,
  "Material": [int]
}
```

```
}
```

Parameter explanation

When a parameter is not provided, the default value will be used

Fixed building parameters

Name	Description	Values	Default	Notes
ConstructionYearCategory	The period in which the house was constructed.	1 = t/m 1924 2 = from 1925 t/m 1945 3 = from 1946 t/m 1964 4 = from 1965 t/m 1974 5 = from 1975 t/m 1991 6 = from 1992 t/m 2005 7 = from 2006 t/m 2014 8 = from 2015	-	Mandatory (CYC)
SubType	The type of apartment	1 = corner ground floor 2 = in between ground floor 3 = corner mid floor 4 = in between mid floor 5 = corner top floor 6 = in between top floor 7 = entire top floor	-	Mandatory
NumberOfStories	The number of stories of the unit	1 = 1 story/building layers 2 = 2 stories/building layers	-	Mandatory
BackFacade	If the apartment has as back facade (achtergevel)	0 = no back facade 1 = back facade	-	Mandatory
LivingArea	The total living area, (gebruiksoppervlak)	Float, > 0.0 in m^2	-	Mandatory, soft limit on min and max

Current envelope and installation parameters

Name	Description	Values	Default	Notes
WallInsulation	The insulation level of the walls.	1 = Geen 2 = Matig 3 = Goed 4 = Zeer Goed	1 if CYC <=4, 2 if CYC <=5, 3 if CYC <=6, 4 if CYC > 6	
FloorInsulation	The insulation level of the floor.	1 = Geen 2 = Matig 3 = Goed 4 = Zeer Goed	1 if CYC <=4, 2 if CYC <=5, 3 if CYC <=6, 4 if CYC > 6	
RoofInsulation	The insulation level of the roof sections.	1 = Geen 2 = Matig 3 = Goed 4 = Zeer Goed	1 if CYC <=4, 2 if CYC <=5, 3 if CYC <=6, 4 if CYC > 6	
GlassLivingArea	The type of glass in the living area.	1 = Single 2 = Double 3 = HR++ 4 = Triple	2 if CYC <=6, 3 if CYC > 6	
Installation	The type of installation used for heating and hot water	1 = Lokaal gas + geiser 2 = VR ketel + geiser 3 = VR combi ketel 4 = HR combi ketel	4	

		6 = Elektrishe lucht-water WP 7 = Stadverwarming 8 = Hybride WP 9 = Collectieve gasketel 10 = Collectieve Warmtepomp		
ShowerHeatRecovery	Is there a shower heat recovery system	1 = No shower heat recovery 2 = Vertical shower heat recovery system	1	
Cooling	Is there a cooling system present	1 = No cooling 2 = air conditioning unit, direct air cooling	1	
Ventilation	The type of ventilation system present	1 = Natuurlijke ventilatie 2 = Mechanische afzuiging 3 = Balansventilatie met wtw 4 = Decentrale balansventilatie	1 if CYC <=4, 2 if CYC <=7, 3 if CyC > 7	
ElectricCooking	What is used for cooking	1 = Gas stove 2 = Electric stove	2 if Installation in [6,7], Else 1	
WallInsulationTarget	The target insulation level of the walls.	1 = Geen 2 = Matig 3 = Goed 4 = Zeer Goed	1 if CYC <=4, 2 if CYC <=5, 3 if CYC <=6, 4 if CYC > 6	
FloorInsulationTarget	The target insulation level of the floor.	1 = Geen 2 = Matig 3 = Goed 4 = Zeer Goed	1 if CYC <=4, 2 if CYC <=5, 3 if CYC <=6, 4 if CYC > 6	
RoofInsulationTarget	The target insulation level of the roof sections.	1 = Geen 2 = Matig 3 = Goed 4 = Zeer Goed	1 if CYC <=4, 2 if CYC <=5, 3 if CYC <=6, 4 if CYC > 6	
GlassLivingAreaTarget	The target type of glass in the living area.	1 = Single 2 = Double 3 = HR++ 4 = Triple	2 if CYC <=6, 3 if CYC > 6	
InstallationTarget	The target type of installation used for heating and hot water	1 = Lokaal gas + geiser 2 = VR ketel + geiser 3 = VR combi ketel 4 = HR combi ketel 5 = HR combi met zonneboiler 6 = Elektrishe lucht-water WP 7 = Stadverwarming 8 = Hybride WP	4	
ShowerHeatRecoveryTarget	Is there a shower heat recovery system target	1 = No shower heat recovery 2 = Vertical shower heat recovery system	1	
CoolingTarget	Is there a cooling system present in the target	1 = No cooling 2 = air conditioning unit, direct air cooling	1	
VentilationTarget	The type of ventilation system present in the target	1 = Natuurlijke ventilatie 2 = Mechanische afzuiging 3 = Balansventilatie met wtw 4 = Decentrale balansventilatie	1 if CYC <=4, 2 if CYC <=7, 3 if CyC > 7	
ElectricCookingTarget	What is used for cookingin the target	1 = Gas stove 2 = Electric stove	2 if Installation in [6,7], Else 1	

User related parameters

Name	Description	Values	Default	Notes
Residents	Number of residents	Int > 0. Values > 6 will be set to 6	3	
GasConsumption	User specified yearly gasconsumption in m^3	Float > 0.0	calculation	
HeatConsumption	User specified yearly heat consumption in GJ.	Float > 0.0	calculation	Only with installation 7
ElectricityConsumption	The gross user specified yearly electricity consumption in KWh	Float, > 0.0	calculatoin	
ElectricityGeneration	The total yearly electricity yield of solar panels in kWh. Net usage = consumption - yield	Float, > 0.0	calculation	

Solar panels

Name	Description	Values	Default	Notes
SolarPanels	A list of solar panel objects, including roof parameters	list	[]	See below

SolarPanels field

Name	Description	Values	Default	Notes
PVTotalWattPeak	The total wattpeak power of the existing system (watt piek). = PVArea * SpecificWattpeak	float, > 0	calculation	When inputting existing system: either PVTotalWattPeak or (PVArea and SpecificWattpeak) is mandatory. 0 if there is no existing system
PVArea	The total effective area of all existing solar panels combined. Standard panel size is 1.65 m^2	Float, > 0.0	1.65 m^2 per panel	When inputting existing system: either PVTotalWattPeak or (PVArea and SpecificWattpeak) is mandatory. 0 if there is no existing system
SpecificWattpeak	The wattpeak/m^2 of the panels. Standard panel size is 1.65 m^2	Float, > 0.0	250 wattpeak/m^2	When inputting existing system: either PVTotalWattPeak or (PVArea and SpecificWattpeak) is mandatory. 0 if there is no existing system
PVTotalWattPeakTarget	The total wattpeak power of the system (watt piek) in target situation. = PVArea * SpecificWattpeak	float, > 0	calculation	When inputting new system: either PVTotalWattPeak or (PVArea and SpecificWattpeak) is mandatory. 0 if there is no existing system
PVAreaTarget	The total effective area of all solar panels combined in the target situation. Standard panel size is 1.65 m^2	Float, > 0.0	1.65 m^2 per panel	When inputting new system: either PVTotalWattPeak or (PVArea and SpecificWattpeak) is mandatory. 0 if there is no

				existing system
SpecificWattpeakTarget	The wattpeak/m^2 of the new panels. Standard panel size is 1.65 m^2	Float, > 0.0	250 wattpeak/m^2	When inputting system: either PVTotalWattPeak or (PVArea and SpecificWattpeak) is mandatory. 0 if there is no existing system
InstallationAngle	The angle of the solar panels	int> 0, <= 90	45	mandatory
Orientation	The orientation of the solar panels	0 = N 45 = NO 90 = O 135 = ZO 180 = Z 225 = ZW 270 = W 315 = NW		mandatory
FreeRoofArea	The free space on this roof surface in m^2 that can be used for additional panels.	Float > 0.0	0	

Financial parameters

Name	Description	Values	Default	Notes
ExcludingVat	Prices are with or without VAT included.	0 = VAT included 1 = VAT excluded	0	Not yet fully implemented
ConstIndicators	ConsIndicator object	json	{}]	See below

CostIndicators field

(standards from cbs may 2025, https://www.cbs.nl/nl-nl/cijfers/detail/85592NED

Name	Description	Values	Default	Notes
fixed_gas_price	Fixed gas cost.Transporttarief + vast leverlingstarief.	Float, > 0.0	337.89	In €/year
variable_gas_price	Variable price exc. tax. Variabel leveringstarief contractprijs	Float > 0.0	0.6595	In €/m3
energy_tax_gas	Energy tax on gas.	Float > 0.0	0.69957	In €/m3
fixed_electricity_price	Fixed electricity price. Transporttarief + vast leveringstarief	Float, > 0.0	575.76	In €/year
variable_electricity_price	Variable price exc. tax. Variabel leveringstarief contractprijs	Float, > 0.0	0.1470	In €/kWh
energy_tax_electricity	Energy tax on electricity.	Float, > 0.0	0.12286	In €/kWh
fixed_heat_price	From Vattenvall, tarifs 2025	Float, > 0.0	760.77	In €/year
energy_tax_heat	Tax on consumed external heat	Float, > 0.0	0	In €/GJ
variable_heat_price	From Vattenvall, including, rent afleverset	Float, > 0.0	43.79	In €/GJ

tax_credit	Vermindering energiebelasting	Float, > 0.0	635.19	In €/year
feed_in_compensation	Earnings per kwh fed back in the grid	Float, > 0.0	0.1331	€/kWh
feed_in_cost	Cost per kwh fed back in the grid	Float, > 0.0	0.125	€/kWh
fixed_solar_panel_cost	Yearly extra charge from electricity supplier if you have solar panels	Float, > 0.0	0.0	€/year
netting_percentage	The fraction of produced electricity that is returned to the grid (and will count towards salderingsregeling.	Float, > 0.0	0.65	
gas_yearly_price_increase	Expected percentage increase in gas price	Float, > 0.0	0.03	
electricity_yearly_price_increase	Expected percentage increase in electricity price	Float, > 0.0	0.02	
external_heating_yearly_price_increase	Expected percentage increase in external heating price	Float, > 0.0	0.03	
term	The term in years used for calculation return on investment	Int, > 0	25	year
discount_rate	Discount rate (~inflation) used in return on investment calculation	Float	0.04	
tax_rate	General VAT	float	1.21	
custom_measure_costs	{“RVO_code” value}	json	{}	To be expanded with filtereable parameters

Output parameters

Name	Description	Values	Default	Notes
ResultsPerMeasure	Whether subresults for each individual measure are shown	0 = don’t show 1 = show	0	

Measure filters

Name	Description	Values	Default	Notes
FloorMeasures	Filters applied to allowable floor measures	json		
WallMeasures	Filters applied to allowable wall measures	json		
RoofMeasures	Filters applied to allowable roof measures	json		

FloorMeasures field

Name	Description	Values	Default	Notes
FloorType	Type of floor (material)	1: wood floor 2: concrete floor	None	
CrawlSpace	Does the house have a crawlspace >isulation underside of floor possible	True: crawlspace False: no crawlspace	None	
Material	List of allowed materials for the insulation	1: mineral wool 2: PIR 3: EPS 4: Resol	None	Input as [1, 2, 3, n]

WallMeasures field

Name	Description	Values	Default	Notes
Biobased	Use biobased materials?	True: only use biobased options False: both biobased and non-biobased	None	
HasCavity	Are cavity walls present suitable for insulation	True: cavity exists and can be filled False: no cavity walls or not suitable	None	
Material	List of allowed materials for the insulation	1: mineral wool 2: PIR 3: EPS pearls 4: Wood fiber 5: PU foam	None	Input as [1, 2, 3, n]

RoofMeasures field

Name	Description	Values	Default	Notes
Biobased	Use biobased materials?	True: only use biobased options False: both biobased and non-biobased	None	
Side	Inside of outside of roof	1: only inside insulation 2: only outside insultation	None	
ReplaceRoof	Consider replacing the roof?	True: Options include ones with roof replacement False: Only options without roof replacement		
Material	List of allowed materials for the insulation	1: PIR 2: Bio EPS 3: EPS 4: XPS 5: Flax 6: Wood fibre 7: Mineral wool 8: Resol	None	

Output

Json output structure


```
{
 "Warnings": [
   {
     "code": "string",
     "description": "string"
   },
 ],
 "Current_House": {
   "Energy_label": str,
   "BENG2": int,
   "Wall_Insulation": int,
   "Floor_Insulation": int,
   "Roof_Insulation": int,
   "Glass_Living_Area": int,
   "Installation": int,
   "Shower_Heat_Recovery": int,
   "Cooling": int,
   "Ventilation": int,
   “ElectricCooking”: int,
   "Solar_Panels": [
     {
       "Roof_Surface": 0,
       "Total_Watt_Peak_Capacity": int,
       "Current_specific_Watt_Peak": float,
       "Inclination_Angle": int,
       "Orientation": int
     },
     {
       "Roof_Surface": n,
       "Total_Watt_Peak_Capacity": int,
       "Current_specific_Watt_Peak": float,
       "Inclination_Angle": int,
       "Orientation": int
     }
   ],
   "CO2_Emission": int,
   "Gross_Electricity_Use_KWh": int,
   "Solar_panel_Yield_KWh": int,
   "Gas_Use_M3": int,
   "External_Heating_Use_Gj": int,
   "Yearly_Energy_Cost": int,
   "Monthly_Energy_Cost": float
 },
```

```
"Estimated_Geometric_Areas": {
  "floor": float,
  "wall": float,
  "roof":float,
  "window_living": float,
  "doors": float
},
"Resulting_House": {
  "Energy_label": "str",
  "BENG2": int,
  "Wall_Insulation": int,
  "Floor_Insulation": int,
  "Roof_Insulation": int,
  "Glass_Living_Area": int,
  "Installation": int,
  "Shower_Heat_Recovery": int,
  "Cooling": int,
  "Ventilation": int,
  “ElectricCooking”: int,
  "Solar_Panels": [
    {
      "Roof_Surface": 0,
      "Total_Watt_Peak_Capacity": int,
      "Inclination_Angle": int,
      "Orientation": int,
      "Added_Watt_Peak_capacity": int,
      “Specific_Watt_Peak_Of_Added_Panels”: int
      "Added_Panels": int,
      "Panel_Size_M2": float,

    },
    {
      "Roof_Surface": n,
      "Total_Watt_Peak_Capacity": int,
      "Inclination_Angle": int,
      "Orientation": int,
      "Added_Watt_Peak_capacity": int,
      “Specific_Watt_Peak_Of_Added_Panels”: int
      "Added_Panels": int,
    },
  ],
  "CO2_Emission": int,
  "Gross_Electricity_Use_KWh": int,
  "Solar_panel_Yield_KWh": int,
  "Gas_Use_M3": int,
  "External_Heating_Use_Gj": int,
  "Yearly_Energy_Cost": int,
  "Monthly_Energy_Cost":int,
```

```
    "Yearly_Energy_Savings": int,
    "Montyly_Energy_Savings": int,
    "Investment_Costs": int,
    "Return_On_Investment": float,
    "CO2_Reduction": int,
    “Variable_Gas_Price”: float,
    “Variabel_Electricity_Price”: float,
  },
  "Results_per_Measure": {
      "<measure name>": {
      "Current_Value": int,
      "New_Value": int,
      "Investment": int,
      “Electricity_saving_kwh”: int,
      “Gas_saving_m3”: int,
      "Energy_Cost_Savings": int,
      "CO2_Reduction": int,
      "Relative_BENG2_Improvement": int
      },
     {
       …
      }
   }
}
```

current apartments

Only current result

Description

This endpoint only returns the result of the current building. The input is a subset of /v2.0. When only the current building results are needed, this endpoint is slightly faster than /v2.0.

Endpoint: /current

Api input json structure ->/current (only current building)

{
"ConstructionYearCategory": int,
"Subtype": int,
"LivingArea": float,
"NumberOfStories": int,
"BackFacade": int,
"WallInsulation": int,
"FloorInsulation": int,
"RoofInsulation": int,
"GlassLivingArea": int,
"Installation": int,
"ShowerHeatRecovery": int,
"Cooling": int,
"Ventilation": int,
"ElectricCooking": int,
"Residents": int,
"GasConsumption": float,
"HeatConsumption": float,
"ElectricityConsumption": float,
"ElectricityGeneration": float,
"SolarPanels": [
{
"PVTotalWattPeak": float,
"PVArea": float,
"SpecificWattpeak":float,
"InstallationAngle": int,
"Orientation": int,
"FreeRoofArea": float,
},
{....}
],

```
  "ExcludingVat": int,
  "CostIndicators": {
    "fixed_gas_price": float,
    "variable_gas_price": float,
    "energy_tax_gas": float,
    "fixed_electricity_price": float,
    "variable_electricity_price": float,
    "energy_tax_electricity": float,
    "fixed_heat_price": float,
    "energy_tax_heat": float,
    "variable_heat_price": float,
    "tax_credit": float,
    "feed_in_compensation": float,
    "feed_in_cost": float,
    "fixed_solar_panel_costs": float,
    "netting_percentage": float,
    "gas_yearly_price_increase": float,
    "electricity_yearly_price_increase": float,
    "external_heating_yearly_price_increase": float,
    "term": int,
    "discount_rate": float,
    "tax_rate": float
  }
}
```

Parameter explanation

When a parameter is not provided, the default value will be used

Fixed building parameters

Name	Description	Values	Default	Notes
ConstructionYearCategory	The period in which the house was constructed.	1 = t/m 1924 2 = from 1925 t/m 1945 3 = from 1946 t/m 1964 4 = from 1965 t/m 1974 5 = from 1975 t/m 1991 6 = from 1992 t/m 2005 7 = from 2006 t/m 2014 8 = from 2015	-	Mandatory (CYC)
SubType	The type of apartment	1 = corner ground floor 2 = in between ground floor 3 = corner mid floor 4 = in between mid floor 5 = corner top floor 6 = in between top floor 7 = entire top floor	-	Mandatory
NumberOfStories	The number of stories of the unit	1 = 1 story/building layers 2 = 2 stories/building layers	-	Mandatory

BackFacade	If the apartment has as back facade (achtergevel)	0 = no back facade 1 = back facade	-	Mandatory
LivingArea	The total living area, (gebruiksoppervlak)	Float, > 0.0 in m^2	-	Mandatory, soft limit on min and max

Current envelope and installation parameters

Name	**Description**	**Values**	**Default**	**Notes**
WallInsulation	The insulation level of the walls.	1 = Geen 2 = Matig 3 = Goed 4 = Zeer Goed	1 if CYC <=4, 2 if CYC <=5, 3 if CYC <=6, 4 if CYC > 6	
FloorInsulation	The insulation level of the floor.	1 = Geen 2 = Matig 3 = Goed 4 = Zeer Goed	1 if CYC <=4, 2 if CYC <=5, 3 if CYC <=6, 4 if CYC > 6	
RoofInsulation	The insulation level of the roof.	1 = Geen 2 = Matig 3 = Goed 4 = Zeer Goed	1 if CYC <=4, 2 if CYC <=5, 3 if CYC <=6, 4 if CYC > 6	
GlassLivingArea	The type of glass in the living area.	1 = Single 2 = Double 3 = HR++ 4 = Triple	2 if CYC <=6, 3 if CYC > 6	
Installation	The type of installation used for heating and hot water	1 = Lokaal gas + geiser 2 = VR ketel + geiser 3 = VR combi ketel 4 = HR combi ketel 6 = Elektrishe lucht-water WP 7 = Stadverwarming 8 = Hybride WP 9 = Collectieve HR ketel 10 = Collectieve warmtepomp	4	
ShowerHeatRecovery	Is there a shower heat recovery system	1 = No shower heat recovery 2 = Vertical shower heat recovery system	1	
Cooling	Is there a cooling system present	1 = No cooling 2 = air conditioning unit, direct air cooling	1	
Ventilation	The type of ventilation system present	1 = Natuurlijke ventilatie 2 = Mechanische afzuiging 3 = Balansventilatie met wtw 4 = Decentrale balansventilatie	1 if CYC <=4, 2 if CYC <=7, 3 if CyC > 7	
ElectricCooking	What is used for cooking	1 = Gas stove 2 = Electric stove	2 if Installation in [6,7], Else 1	

User related parameters

Name	**Description**	**Values**	**Default**	**Notes**

Residents	Number of residents	Int > 0. Values > 6 will be set to 6	3	
GasConsumption	User specified yearly gasconsumption in m^3	Float > 0.0	calculation	
HeatConsumption	User specified yearly heat consumption in GJ.	Float > 0.0	calculation	Only with installation 7
ElectricityConsumption	The gross user specified yearly electricity consumption in KWh	Float, > 0.0	calculatoin	
ElectricityGeneration	The total yearly electricity yield of solar panels in kWh. Net usage = consumption - yield	Float, > 0.0	calculation	

Solar panels

Name	Description	Values	Default	Notes
SolarPanels	A list of solar panel objects, including roof parameters	list	[]	See below

SolarPanels field

Name	Description	Values	Default	Notes
PVTotalWattPeak	The total wattpeak power of the existing system (watt piek). = PVArea * SpecificWattpeak	float, > 0	calculation	When inputting existing system: either PVTotalWattPeak or (PVArea and SpecificWattpeak) is mandatory. 0 if there is no existing system
PVArea	The total effective area of all existing solar panels combined. Standard panel size is 1.65 m^2	Float, > 0.0	1.65 m^2 per panel	When inputting existing system: either PVTotalWattPeak or (PVArea and SpecificWattpeak) is mandatory. 0 if there is no existing system
SpecificWattpeak	The wattpeak/m^2 of the panels. Standard panel size is 1.65 m^2	Float, > 0.0	250 wattpeak/m^2	When inputting existing system: either PVTotalWattPeak or (PVArea and SpecificWattpeak) is mandatory. 0 if there is no existing system
InstallationAngle	The angle of the solar panels	int> 0, <= 90	45	mandatory
Orientation	The orientation of the solar panels	0 = N 45 = NO 90 = O 135 = ZO 180 = Z 225 = ZW 270 = W 315 = NW		mandatory
FreeRoofArea	The free space on this roof surface in m^2 that can be used for additional panels.	Float > 0.0	0	

Financial parameters

Name	Description	Values	Default	Notes
ExcludingVat	Prices are with or without VAT included.	0 = VAT included 1 = VAT excluded	0	Not yet fully implement ed
ConstIndicators	ConsIndicator object	json	{}]	See below

CostIndicators field

(standards from cbs may 2025, https://www.cbs.nl/nl-nl/cijfers/detail/85592NED

Name	Description	Values	Default	Notes
fixed_gas_price	Fixed gas cost.Transporttarief + vast leverlingstarief.	Float, > 0.0	337.89	In €/year
variable_gas_price	Variable price exc. tax. Variabel leveringstarief contractprijs	Float > 0.0	0.6595	In €/m3
energy_tax_gas	Energy tax on gas.	Float > 0.0	0.69957	In €/m3
fixed_electricity_price	Fixed electricity price. Transporttarief + vast leveringstarief	Float, > 0.0	575.76	In €/year
variable_electricity_price	Variable price exc. tax. Variabel leveringstarief contractprijs	Float, > 0.0	0.1470	In €/kWh
energy_tax_electricity	Energy tax on electricity.	Float, > 0.0	0.12286	In €/kWh
fixed_heat_price	From Vattenvall, tarifs 2025	Float, > 0.0	760.77	In €/year
energy_tax_heat	Tax on consumed external heat	Float, > 0.0	0	In €/GJ
variable_heat_price	From Vattenvall, including, rent afleverset	Float, > 0.0	43.79	In €/GJ
tax_credit	Vermindering energiebelasting	Float, > 0.0	635.19	In €/year
feed_in_compensation	Earnings per kwh fed back in the grid	Float, > 0.0	0.1331	€/kWh
feed_in_cost	Cost per kwh fed back in the grid	Float, > 0.0	0.125	€/kWh
fixed_solar_panel_cost	Yearly extra charge from electricity supplier if you have solar panels	Float, > 0.0	0.0	€/year
netting_percentage	The fraction of produced electricity that is returned to the grid (and will count towards salderingsregeling.	Float, > 0.0	0.65	
gas_yearly_price_increase	Expected percentage increase in gas price	Float, > 0.0	0.03	
electricity_yearly_price_incre ase	Expected percentage increase in electricity price	Float, > 0.0	0.02	
external_heating_yearly_pric e_increase	Expected percentage increase in external heating price	Float, > 0.0	0.03	
term	The term in years used for calculation	Int, > 0	25	year

	return on investment			
discount_rate	Discount rate (~inflation) used in return on investment calculation	Float	0.04	
tax_rate	General VAT	float	1.21	

Output

Json output structure

```
{
 "Warnings": [
  {
   "code": "string",
   "description": "string"
  },
  {
   "code": "01",
   "description": "string"
  }
 ],
 "Current_House": {
  "Energy_label": str,
  "BENG2": int,
  "Wall_Insulation": int,
  "Floor_Insulation": int,
  "Roof_Insulation": int,
  "Glass_Living_Area": int,
  "Installation": int,
  "Shower_Heat_Recovery": int,
  "Cooling": int,
  "Ventilation": int,
  “ElectricCooking”: int,
  "Solar_Panels": [
   {
    "Roof_Surface": 0,
    "Total_Watt_Peak_Capacity": int,
    "Inclination_Angle": int,
    "Orientation": int
   },
   {
    "Roof_Surface": n,
    "Total_Watt_Peak_Capacity": int,
    "Inclination_Angle": int,
    "Orientation": int
```

```
      }
    ],
    "CO2_Emission": int,
    "Gross_Electricity_Use_KWh": int,
    "Solar_panel_Yield_KWh": int,
    "Gas_Use_M3": int,
    "External_Heating_Use_Gj": int,
    "Yearly_Energy_Cost": int,
    "Monthly_Energy_Cost": float
  },
  "Estimated_Geometric_Areas": {
    "floor": float,
    "wall": float,
    "roof":float,
    "window_living": float,
    "doors": float
  }
}
```

Appendix A: Custom measure costs

Appendix A: Custom measure costs

Only insulation measures (floor, wall, roof) have filters. In case (part of) a measure is not defined in the list from RVO, a custom code is provided with description for that specific measure. Costs in €, excluding VAT.

At this time, it is only possible for the user to input custom costs for the measures already defined below. It is not possible to input a complete new measure, including the filters and conditionals.

Floor

category	RVO code	Custom code	description	unit	unit cost
floor	WB002a		Vloerisolatie (Rc=2,1): minerale wol isolatie (d=90mm, Rd=2,4) onderzijde houten begane grondvloer	m²	49
floor	WB002b		Vloerisolatie (Rc=2,6): minerale wol isolatie (d=90mm, Rd=2,4) onderzijde steenachtige begane grondvloer	m²	30
floor	WB002c		Vloerisolatie (Rc=2,1): EPS isolatieplaten (d=100mm) onderzijde houten begane grondvloer	m²	54
floor	WB002d		Vloerisolatie (Rc=2,9): EPS isolatieplaten (d=100mm, Rd=2,8) onderzijde steenachtige begane grondvloer	m²	33
floor	WB002f		Vloerisolatie (Rc=3,7): PIR isolatie (d=100mm) onderzijde houten begane grondvloer	m²	52
floor	WB143		Vloerisolatie (Rc=3,7): PIR isolatie (d=90mm) onderzijde steenachtige begane grondvloer	m²	52
floor	WB001e		Vloerisolatie (Rc=3,1): resol isolatie (d=65mm) bovenzijde houten begane grondvloer - afwerking plaatmateriaal	m²	109

filters

RVO code	biobased	floor_type	material	crawlspace	Version
WB002a	FALSE	wood	mineral_wool	TRUE	v1
WB002b	FALSE	concrete	mineral_wool	TRUE	
WB002c	FALSE	wood	EPS	TRUE	v1
WB002d	FALSE	concrete	EPS	TRUE	
WB002f	FALSE	wood	PIR	TRUE	v1
WB143	FALSE	concrete	PIR	TRUE	
WB001e	FALSE	wood, concrete	resol	FALSE	

Allowed measures

from/to	1	2	3	4
1	-		WB002a, WB002b, WB001e	WB002f, WB143
2		-		WB002c, WB002d, WB001e
3			-	WB002c, WB002d
4				-

Wall

category	RVO code	Custom code	description	unit	unit cost
wall	WB009a		Spouwisolatie (Rc=1,8): minerale wol vlokken (d=50mm, Rd=1,4)	m²	14
wall	WB009b		Spouwisolatie (Rc=1,6): EPS parels (d=50mm)	m²	18
wall	WB009c		Spouwisolatie (Rc=2,1): PU schuim (d=50mm)	m²	40
wall	WB207		Gevelisolatie (Rc=4,2): PIR isolatie (d=100mm, Rd=4,5) binnenzijde gevel - houten regelwerk en gipsbeplating - behangklaar	m²	156
wall	WB242		Gevelisolatie (Rc=2,9): PIR isolatie (d=70mm, Rd=3,15) binnenzijde gevel - houten regelwerk en gipsbeplating - behangklaar	m²	136
wall	WB268		Gevelisolatie (Rc= 2,2) Minerale wol (d=70mm, Rd=2,0) - binnenzijde gevel - houten regelwerk en gipsbeplating - behangklaar	m²	137
wall	WB361		Gevelisolatie (Rc=1,9): Houtvezel (d=60mm, Rd=1,6) - binnenzijde gevel- houten regelwerk en gipsbeplating - behangklaar	m²	132
wall	WB362		Gevelisolatie (Rc=2,8): Houtvezel (d=100mm, Rd=2,6) - binnenzijde gevel- houten regelwerk en gipsbeplating - behangklaar	m²	140
wall	WB363		Gevelisolatie (Rc=3,7): Houtvezel (d=140mm, Rd=3,7) - binnenzijde gevel- houten regelwerk en gipsbeplating - behangklaar	m²	149
wall	WB392		Spouwisolatie (Rc=1,5): minerale wol vlokken (d=40mm, Rd=1,1)	m²	12

filters

RVO code	biobased	wall_type	material	version
WB009a	FALSE	cavity	mineral_wool	
WB009b	FALSE	cavity	EPS_pearls	v1
WB009c	FALSE	cavity	PU_foam	
WB207	FALSE	inside	PIR	
WB242	FALSE	inside	PIR	v1
WB268	FALSE	inside	mineral_wool	
WB361	TRUE	cavity	wood_fibre	v1

WB362	TRUE	inside	wood_fibre	v1
WB363	TRUE	inside	wood_fibre	v1
WB392	FALSE	cavity	mineral_wool	

Allowed measures

*For wall insulation, the option for cavity insulation depends on cavity thickness, which is dependent on construction year. For this reason, separate tables are given per ConstructionYearCategory (CYC).

CYC <= 3

from/to	1	2	3	4
1	-	WB361	WB242, WB362	WB207, WB363
2		-	WB361	WB242, WB362
3			-	WB242, WB361
4				-

CYC = 4

from/to	1	2	3	4
1	-	WB361, WB392	WB009c, WB362	WB009a+WB268, WB009b+WB242, WB363
2		-	WB361	WB242, WB362
3			-	WB242, WB361
4				-

CYC = 5

from/to	1	2	3	4
1	-	-	-	-
2		-	WB009a, WB009b, WB361	WB009a+WB268, WB362
3			-	WB242, WB361
4				-

CYC = 6

from/to	1	2	3	4
1	-	-	-	-
2	-	-	-	-
3			-	WB242, WB361
4				-

Roof

category	RVO code	Custom code	description	unit	unit cost
roof	WB006a		Dakisolatie (Rc=3,2): EPS isolatie (d=100mm) op bestaande dakbedekking plat dak - ballastlaag hergebruiken	m²	236
roof	WB007c		Dakisolatie (Rc=3,7): PIR isolatie (d=80mm) buitenzijde plat dak - vervangen dakbedekking APP	m²	282
roof	WB211		Dakisolatie (Rc=6,3): PIR isolatie (d=160mm) buitenzijde plat dak - vervangen dakbedekking APP	m²	303
roof	WB232		Dakisolatie (Rc=3,0): XPS isolatie (d=80mm, Rd=2,3) op bestaande dakbedekking plat dak - ballastlaag hergebruiken, exclusief dakrand vervangen op matig geisoleerd dak	m²	81
roof	WB265		Dakisolatie (Rc=4,4): XPS isolatie (d=140mm, Rd=4,1) op bestaande dakbedekking plat dak – ballastlaag hergebruiken, exclusief dakrand vervangen op ongeisoleerd dak	m²	91
roof	WB373		Dakisolatie (Rc=5,2): Bio EPS (d=160mm Rd=5,0) buitenzijde plat dak - vervangen dakbedekking APP	m²	287
roof	WB374		Dakisolatie (Rc=6,5): Bio EPS (d=200mm, Rd=6,3) buitenzijde plat dak - vervangen dakbedekking APP	m²	295

filters

RVO code	biobased	roof_type	side	replace_roof	material	version
WB006a	FALSE	flat	outside	FALSE	EPS	
WB007c	FALSE	flat	outside	FALSE	XPS	v1
WB211	FALSE	flat	outside	TRUE	PIR	v1
WB232	FALSE	flat	outside	FALSE	XPS	v1
WB265	FALSE	flat	outside	FALSE	XPS	
WB373	TRUE	flat	outside	TRUE	bio_EPS	
WB374	TRUE	flat	outside	TRUE	bio_EPS	

Allowed measures

For roofs, a distinction is made between sloped and flat roofs.

Flat roof

from/to	1	2	3	4
1	-	WB007c	WB006a, WB007c, WB373, WB232	WB374, WB211
2		-	WB007c, WB232	WB211, WB265, WB374
3			-	WB006a, WB211, WB232, WB374
4				-

Glass

category	RVO code	Custom code	description	unit	unit cost
glass	WB019a		HR++ glas (Ug=1,1) i.p.v. enkel glas	m²	188
glass	WB019b		HR++ glas (Ug=1,1) i.p.v. (standaard) dubbelglas	m²	194
glass	WB161a		Houten kozijn met triple glas (Uw=1,0, Ug=0,6) i.p.v. bestaand houten kozijn met enkel glas	m²	775
glass	WB147b		Triple glas (Ug=0,7) i.p.v. dubbel glas	m²	242
glass	WB147c		Triple glas (Ug=0,7) i.p.v. HR++ glas	m²	242

Allowed measures

from/to	1	2	3	4
1	-		WB019a	WB161a
2		-	WB019b	WB147b
3			-	WB147c
4				-

Installations

category	RVO code	Custom code	description	unit	unit cost
installation	WB042a		Combiketel HR-107 i.p.v. VR-ketel en keukengeiser	-	5227
installation	WB042b		Combiketel HR-107 i.p.v. VR combiketel	-	5653
installation	WB065		Paneelradiatoren met convector T=35-55 i.p.v. paneelradiatoren	-	5379
installation	WB102c		Individuele bemetering warmtemeter aanbrengen	-	585
installation	WB108a		HR-107 combiketel i.p.v. gevelkachels en keukengeisers	-	14272

installation	WB112		Warmtepomp lucht combi en warmtepompboiler i.p.v. gevelkachels en keukengeiser	-	26842
installation	WB131		Aansluiting op warmtenet i.p.v. VR-ketel en keukengeiser	-	4803
installation	WB136		CV-Zonneboiler, voorraadvat 110L, collectoren 4,74 m2, op bestaande CV-ketel, schuin dak	-	4765
installation	WB138		CV-Zonneboiler, voorraadvat 110L, collectoren 4,74 m2, en nieuwe HR107 combiketel, schuin dak, i.p.v. bestaande CV-ketel+keukengeiser	-	8874
installation	WB149a		Warmtepomp lucht combi - Buitenopstelling (ind.) i.p.v. VR-ketel en keukengeiser	-	14348
installation	WB149b		Warmtepomp lucht combi - Buitenopstelling (ind.) i.p.v. HR-combiketel	-	12795
installation	WB149c		Warmtepomp lucht combi - Buitenopstelling (ind.) i.p.v. VR-combiketel	-	15092
installation	WB157		HR-107 combiketel met zonneboiler, voorraadvat 120L, collectoren 2,5 m2 i.p.v. VR combiketel	-	7001
installation	WB159		Hybride warmtepomp en HR-ketel i.p.v. VR-ketel	-	10704
installation	WB198		Hybride warmtepomp i.c.m. HR-ketel i.p.v. gevelkachels en keukengeiser	-	18335
installation	WB199		Hybride warmtepomp bijplaatsen bij HR107-combi ketel	-	5472
installation		IN001	Verwijderen oude lokale installatie, aanpassen leidingen		9031

Allowed measures

from/ to	1	2	3	4	5	6	7	8	9	10
1	-			WB108a	WB108a+WB136	WB112	IN001+WB131+WB102c	WB198		
2		-		WB042a	WB138	WB149a+WB065	WB131+WB102c	WB159		
3			-	WB042b	WB157	WB149c+WB065	WB131+WB102c	WB159		
4				-	WB136	WB149b+WB065	WB131+WB102c	WB199		
5					-	WB149b+WB065	WB131+WB102c	WB199		

6						-				
7							-			
8						WB149b		-		
9									-	
10										-

Douche WTW

category	RVO code	Custom code	description	unit	unit cost
douchewtw	WB142a		Douche WTW	-	666
douchewtw	WB260		Aanbrengen douchegoot WTW	-	2197

Ventilation

category	RVO code	Custom code	description	unit	unit cost
ventilation	WB405		D5b: Decentrale balansventilatie, CO2 gestuurd met dubbelkanaals gevel-WTW in woonkamer en slaapkamer ipv natuurlijke ventilatie	-	6798
ventilation	WB418		D5c: WTW CO2 gestuurd met overflowsysteem in trapgat, i.p.v. mechanische ventilatie	-	7620

Allowed measures

from/to	1	2	3	4
1	-			WB405
2		-	WB418	
3			-	
4				-

Cooling

category	RVO code	Custom code	description	unit	unit cost
cooling		KO001	Single split arco 3.5 kW		1942

Solar panels

category	RVO code	Custom code	description	unit	unit cost
pv	WB234		PV-panelen monokristallijn serieel geschakeld (225Wp/m2, 435Wp/paneel) op hellend dak	m^2	222
pv	WB406		PV-panelen monokristallijn parallel geschakeld (225Wp/m2, 435Wp/paneel) op plat dak	m^2	248

Electric cooking

category	RVO code	Custom code	description	unit	unit cost
stove		ES001	Electric stove	-	1240

Appendix B: V1 Custom cost measures

Appendix B: Custom Measure Cost from v1

The mapping from the old codes to the new codes is found in the table below:

Old code	New code	Filters
10001	WB361	Version: v1
10002	WB362, WB009b	Version: v1
10003	WB242, WB363	Version: v1
30002	WB007c, WB232	Version: v1
30003	WB211	Version: v1
40002	WB005	Version: v1
40003	WB145	Version: v1
50002	WB019a	
50003	WB161a	
50102	WB019b	
50103	WB147b	
50203	WB147c	
60003	WB405	(Amount per house)
80004	WB108a	
80104	WB042a	
80204	WB042b	
80008	WB112	
80108	WB149a+WB065=0	
80408	WB149b+WB065=0	
80009	Geen wkk mogelijk	
80109	Geen wkk mogelijk	
80209	Geen wkk mogelijk	
80030	WB198	
80130	WB159	

80430	WB199	
70000	WB234, WB406	

A couple of things are of note:

1)
An extra measure filter for floors, roofs and walls is added to force the use of the code in the mapping table. So in order to use the mapping table, each request should include:

```
 "FloorMeasures": {
   "Version": "v1"
  },
  "WallMeasures": {
   "Version": "v1"
  },
  "RoofMeasures": {
   "Version": "v1"
  }
```

2)
When there are 2 codes in the “New code” column, both have to be set to the required price. The reason is that in v2, often the current_level of the insulation matters for the measure to be picked. This was not the case in V1.

3)
For full electric heatpump prices, WB149a needs to be set to the required price and WB065 always to 0.

4)
V2 does not have a WKK installation. So measures 80009, 80109 and 80209 can not be input.

5)
Custom costs are input as follows:

```
"CostIndicators": {
	"custom_measure_costs": {
		“WB361”: 35,
		"WB362": 116,
		“WB009b”: 116,
		“WB242”: 156,
		“WB363”: 156,
		etc…
	}
},
```

6)

V2 only has one price to set. So take Materiaal + Werk to calculate. You can ignore Materiaal besparing and Werkbesparing.